Ahsan Mughal

Software Engineer | **PHP / Laravel / MySQL**

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Summary

Backend PHP developer with hands-on experience across both **Laravel** product development and large **legacy PHP/MySQL** systems. Builds secure, scalable web applications and **RESTful APIs**; integrates **Stripe**, **OAuth** and anti-bot protection; and tunes **MySQL** for multi-million-row reporting workloads. Experienced in **role-based access control**, **query optimization**, **database design**, **automated testing** and **CI/CD** within Agile teams, with a focus on clean code, performance and measurable results.

Technical Skills

Languages: PHP, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3/SCSS, C++

Frameworks & Libraries: Laravel 11, Blade, Eloquent ORM, Laravel Breeze, jQuery, Bootstrap, Tailwind CSS, React 19, Next.js 15, Chart.js, Plotly.js, Google Charts

Databases: MySQL, schema design, migrations, indexing & query optimization, stored procedures, triggers, transactions, aggregation/rollup tables

APIs & Integrations: RESTful APIs, AJAX, Stripe (payments & webhooks), Google OAuth, hCaptcha, SMTP/email notifications, webhook handling

Architecture & Practices: MVC, modular & multi-tenant architecture, RBAC & authorization, authentication & session management, caching, legacy system modernization, secure coding (SQL injection prevention, input validation), automated testing & regression suites, code review, Agile/Scrum

DevOps & Deployment: CI/CD pipelines, GitHub Actions (lint, type-check, build gates), Docker & containerized development environments, Git branching & code review workflows, environment configuration (.env, secrets), database migrations, staging & production releases, Vercel, shared hosting/VPS deployment

Tools & Platforms: Git, GitHub, Docker, Composer, npm, Postman, phpMyAdmin, DBeaver, VS Code, Linux, Apache

Experience

Software Engineer (PHP/MySQL), Bristol Mayer Biotech Pakistan – Lahore | bristol.pk — May 2026 – Present

- **Rebuilt the Sales & Reporting suite** of a pharmaceutical ERP (Sales Report, Primary/Secondary/Primary-vs-Secondary Sales Trend, Distributor Trend, Sales Target, Target-vs-Achievement) over **2M+ invoice rows**: a **trigger-synced day-grain rollup table** (2.0M rows / 258MB reduced to 700K rows / 40MB, parity-verified with zero drift), composite and covering indexes and a result cache cut the heaviest report query from **3.9s to 0.7s** and an index-blocking lookup from **15.9s to 0.27s**; delivered multi-level drilldowns, KPI cards, charts, A4 print and Excel export.
- Designed a **materialized territory-sales fact table** that moved incentive calculation from hand-typed figures to real distributor invoices – rebuilt per month inside a **transaction under a named lock**, with **staleness detection**, nightly **cron** rebuilds and a **reconciliation report** accounting for every unattributed unit – verified with **0 mismatches across 42 group-months** before switching the growth-based **incentive engine** and payout lock onto it.
- Designed the **role-based data-scoping model** enforced across every sales screen and dashboard – a 5-level territory hierarchy (MR / RSM / SM / BM / NSM), product-group locking and a dual-tenant ID pattern for multi-company accounts – closing **data-visibility leaks** (a rep's products 42 to 23, distributors 61 to 5) and auditing all **306 hierarchy logins**; rebuilt the **field-force dashboard** as a single JSON payload, cutting a 37s query to a **0.5–1.6s** page load by aggregating before the hierarchy join.
- Built the **Doctor Activity Requisition & Approval system** end to end – 6-step wizard, **multi-level approval routing resolved from the org chart**, approvals inbox, in-place approver edits, printable A4 voucher and recorded payment – with a **status engine**, **immutable line-item snapshots written in a single transaction**, **row locking with conditional updates** against double decisions, a **database-lock critical section** closing a voucher-numbering **race condition**, server-side amount re-derivation blocking tampering, and a **sales-linked cost mode** computing payable amounts from credited pharmacy sales with monthly projection and caps; verified by **250+ automated checks**.

- Built a **Doctor Portal** with **invite-code self-signup** (CSPRNG codes stored as SHA-256 hashes, expiry, attempt limits and per-IP throttling), **bcrypt** passwords, **CSRF tokens** and rate limiting; doctors submit conference, clinic and CME requests through an approval workflow, propose changes to their product and pharmacy lists for manager approval, and track credited sales against expected results.
- Built the **Organogram** and **Team Division** modules – the system's only write interface for assigning and unassigning a territory's full reporting chain, with region-scoped candidate pickers, server-side division locks and an **application-level dual write** keeping territory and profile hierarchies in sync – plus divisional headcount analytics and **4 data-integrity reports** (orphan reps, reps without territories, staff without divisions, cross-region managers).
- Built **Manual Sale Entry** with a **two-stage approval chain read from the org chart** and idempotent recompute-based merging, a shared **notifications** module with de-duplication, and the **Quotation Letter** module (**transactional save**, snapshotted price lines, auto-numbering, isolated-iframe A4 print with a repeating letterhead); shipped every schema change as an **idempotent SQL migration**.

Junior Laravel Developer, TetraLogicx – Lahore Jan 2025 – Apr 2026

- Delivered a **multi-tenant** car rental management system in **Laravel 11** serving **4 user roles** (Admin, Company, Employee, Customer) under **role-based access control**, covering the full booking lifecycle – search, cart, reservation, **Stripe** checkout and commission-based revenue tracking – with **multi-currency and localization** support, advanced vehicle search/filtering, **automated email notifications**, and **analytics dashboards** onboarding **10+ rental companies** and processing **500+ bookings**.
- Built a **Laravel** electronic voting platform with **RBAC** and **3 sign-in paths** (email/password, **Google OAuth**, **email verification**), a multi-step event-creation wizard, a public voting interface with **token-based access**, **Stripe** payments, booking/tariff management, a reward system, **hCaptcha** bot protection, **session management**, **timezone handling** and email notifications; ran **10+ voting events** totalling **1,000+ votes** with zero double-vote incidents.
- Developed a **full-stack Laravel 11** application for a private equity firm – public website plus secure admin panel – with **CRUD operations**, **authentication and authorization** (**Laravel Breeze**), file uploads, email notifications, **Eloquent ORM** data management and a responsive Bootstrap UI, shipped to production in **6 weeks**.
  URL: `https://ascendexits.com/`
- Maintained **Almiranta Cap-Ex Plan**, a **production-grade LAAM-stack application** with **modular architecture**, delivering **CapEx workflows** (budgeting, tracking, dashboards, **cron/export automation**) with **multi-view analysis** and customizable UI logic across **10+ cost centres**, resolving **25+ production issues** against live client data.
  URL: `https://www.capexplan.com/`
- Owned **deployment and release** for these applications: managed **local, staging and production** environments, **environment-variable and secrets configuration**, **database migrations** against live data, and **GitHub Actions CI** pipelines running lint, type-check and build gates on every push and pull request.

Projects

Sentiment Analysis of Pakistan General Elections 2024 github.com/Ahasn-9

- Developed a responsive **election analysis dashboard** using JavaScript, HTML5, CSS3, Chart.js and Plotly.js with 8+ interactive visualizations, integrating a **FastAPI** backend via the Fetch API to process **100K+ tweets** asynchronously using modular, **event-driven JavaScript** components.

Education

COMSATS University Lahore, BS in Software Engineering Feb 2021 – Dec 2024

- **Coursework:** Data Structures and Algorithms (C++), Requirements Analysis and System Architecture, Web Technologies, Discrete Structures, Software Project Management, Software Testing